Ms01: Introduction

Mobile Multi Sensor Systems

Transformation (translation & rotation)

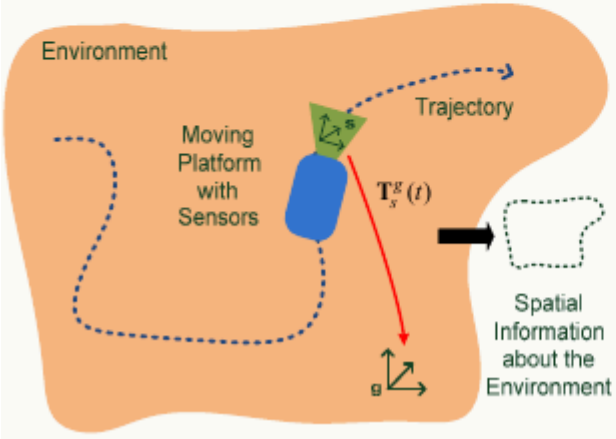
➔ matrices, vectors, quaternion, homogeneous coordinates, ...

global coordinate systems

$$\mathbf{T}_s^g(t)$$

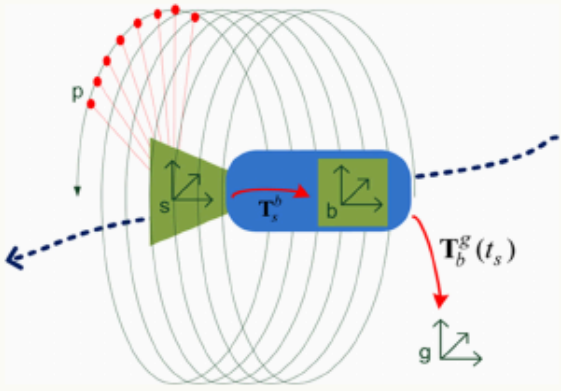
time of the measurement

sensor coordinate systems



Object point in the global coordinate system

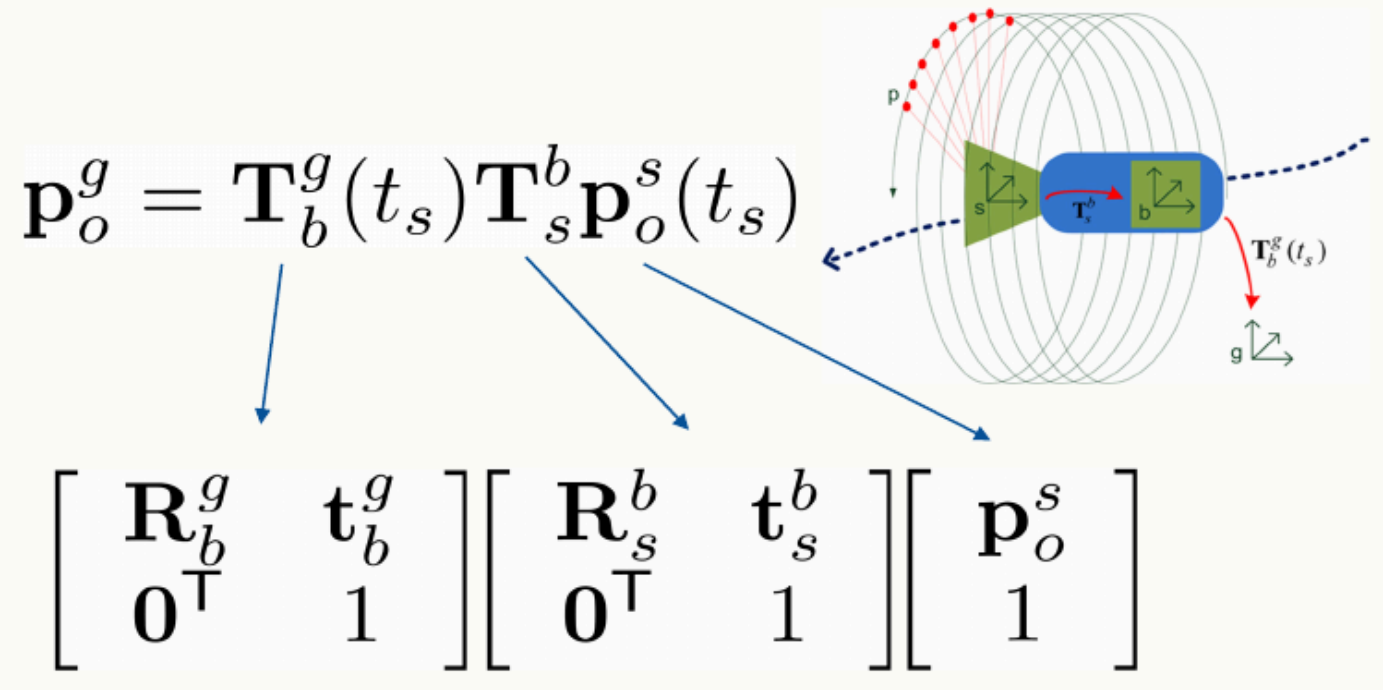
$$\mathbf{p}_o^g = \mathbf{T}_b^g(t_s)\mathbf{T}_s^b\mathbf{p}_o^s(t_s)$$



Object point in Sensor coordinate system (➔ Mapping sensors)

Transformation between Sensor coordinate system and vehicle system (➔ System calibration)

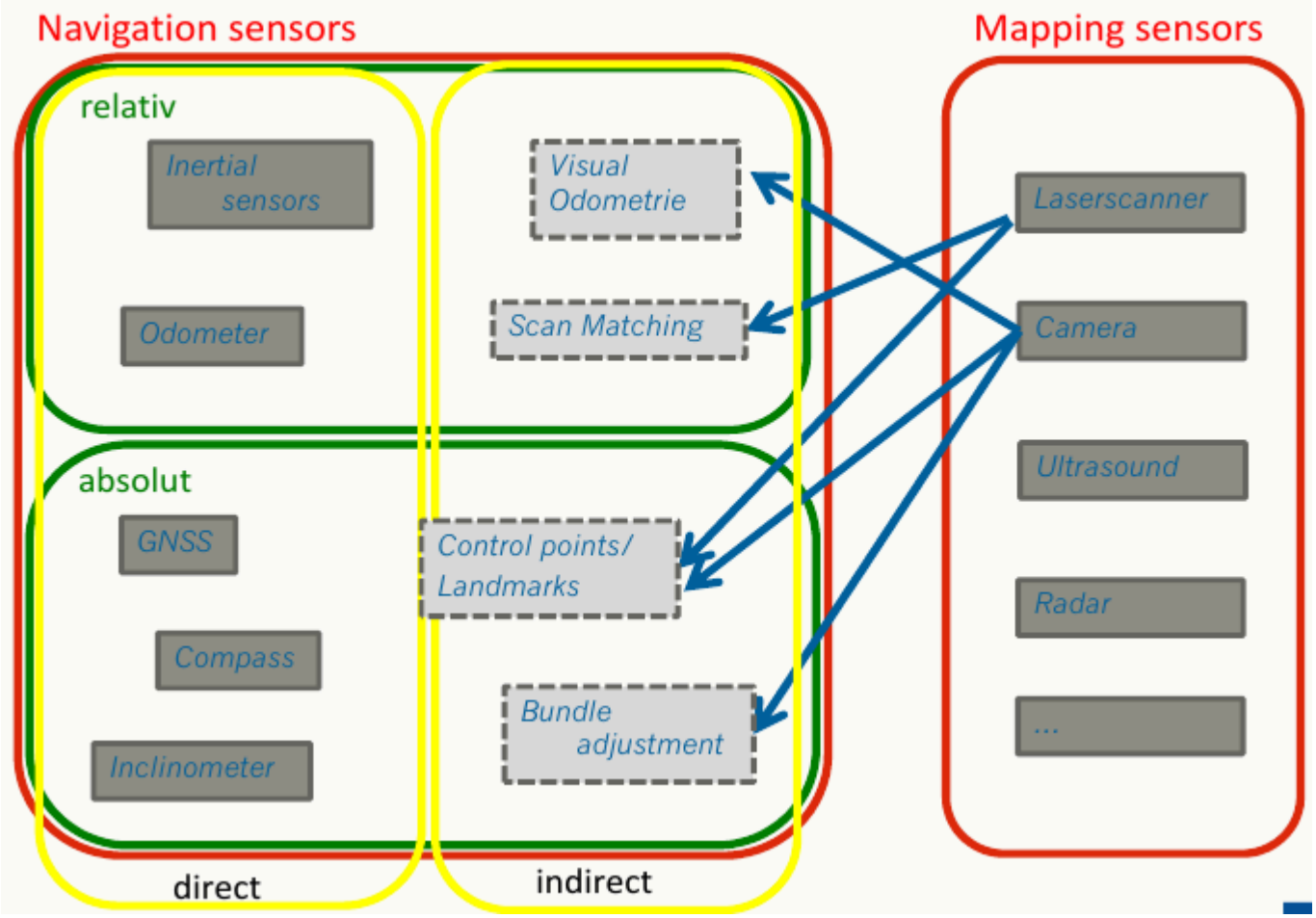
Pose of the vehicle at the time of mapping sensor measurement in the global coordinate system (➔ Trajectory estimation)



Trajectory estimation

$$T_b^g(t)$$

- Estimation of 6 parameters (position and rotation) at each relevant point in time
- **Problem**
  - High accuracy via overdetermination not possible
  - Platform motion disables multiple measurements
- -> Solution
  - Usage of various partly redundant sensors
  - Usage of knowledge about the system (motion models)
  - Usage knowledge about the environment (e.g. maps)
- **Methods**
  - Sensor Fusion
  - Recursive Bayes estimation
  - Kalman Filter, non- linear extension, Smoothing
  - Particle Filter, MonteCarlo methods
  - Bundle adjustment, SLAM



- **Inertial Sensors**
  - Acceleration and angular rate with respect to the inertial system
  - -> relative position and orientation
- **Odometer**
  - Path increments or speed
    - -> dead reckoning
- **GNSS**
  - Absolute position
    - (~m or ~cm, code or carrier phase)
- **Compass**
  - Orientation with respect to magnetic North
- **Inclinometer**
  - Orientation with respect to gravity

Algorithms

- Real-Time Processing
  - Recursive Bayes-Filter

        - EKF /KF
        - Particle filter
- Postprocessing - One adjustment using ALL data
    - Bundle adjustment (example SfM)
        - Simultaneous estimation of a camera pose and object points using corresponding image features
        - Inclusion of GNSS is possible
    - SLAM - Simultaneous Localisation and Mapping
        - Laser scanner or camera
        - Using loop closures

Quality Analysis

- Object point in the global coordinate system -> Accuracy?
- Ideal case: Full covariance matrix of every object point, but
    - High number of measurements and calibration parameters
    - Complex and partly unknown functional model
    - Not normally distributed errors
- Comparison with reference Data!
    - Detection of control points in the scans and comparison with reference coordinates
    - Comparison of kinematic laser scanning point clouds and comparison with TLS scans or 3D models
- -> Difficult to trace back to the error source